

Growth & Remodeling of Soft Tissue

A hands-on introduction to the four modelling frameworks

kinematic growth · constrained mixture · homogenized · equilibrated

teacher version — key equations filled in

Reproducible figures generated by the `gr` package — github.com/YaleCBL-Teaching/Growth-Remodeling

The biology

Living Materials Are Never “Finished”

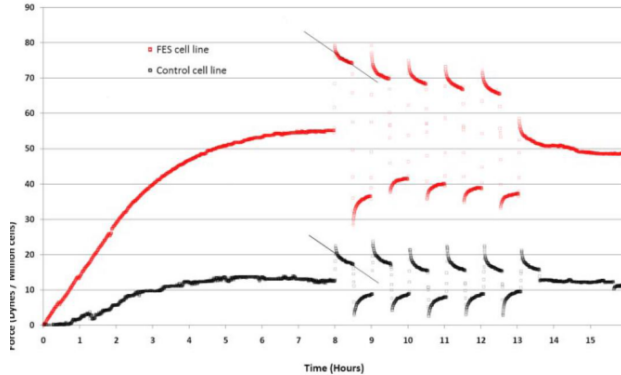
A steel beam is made once and wears out. A living artery is **continuously torn down and rebuilt**: cells deposit and degrade structural proteins while sensing the loads they carry.

- **Growth** — change in *mass*.
- **Remodeling** — change in *microstructure* (fiber orientation, natural length, composition).

Will a hypertensive artery thicken and stabilise, or an aneurysm dilate to rupture?

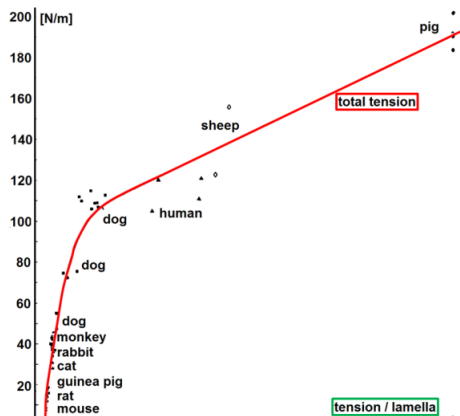
Cells Actively Hold a Mechanical Set-Point

Cultured cells generate a steady **tension** and *restore* it after each perturbation — direct evidence that living tissue targets a mechanical state rather than a fixed shape.



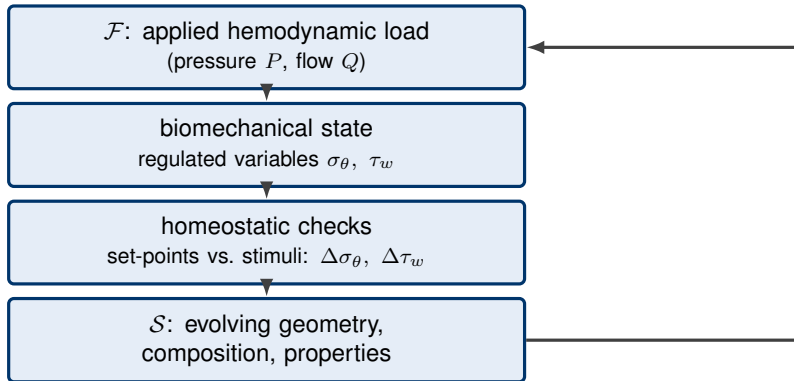
Fundamental Observation: A Stress Set-Point

Wolinsky & Glagov (1967): Cauchy stress in the aortic media is nearly **constant across mammals** spanning orders of magnitude in body mass — tissue-level evidence of a mechanical set-point.



Cells Chase a Mechanical Set-Point

Cells regulate a **target (homeostatic) stress**: above it they build more, below it removal wins — a negative feedback that thickens a hypertensive wall until stress returns to normal.



Two Canonical Scenarios We Will Simulate

Hypertension — sustained pressure step.
Expect *adaptation*: the wall thickens, stress returns to set-point.

Aneurysm — irreversible elastin loss. Geometry can feed *positively* back on stress: stabilise larger, *or* dilate without bound.

When does a tissue adapt, and when does it run away?

The Cast of Characters in an Arterial Wall

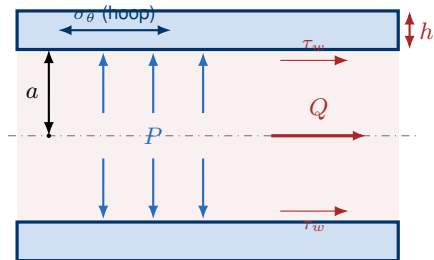
Constituent	Mechanical role	Mass frac. ϕ	Dep. stretch G	Material model & turnover $1/k_d$
Elastin	soft; bears load at low stretch	0.30	1.40	neo-Hookean ($c \approx 90$ kPa); none (decades)
Collagen	stiff, crimped; guards against rupture	0.35	1.08	Fung ($c_1 \approx 250$ kPa, $c_2 \approx 8$); fast (~ 20 d)
Smooth muscle	passive load + active contraction; the sensor cell	0.35	1.20	Fung ($c_1 \approx 120$ kPa, $c_2 \approx 3$); fast (~ 20 d)

- **Elastin cannot be renewed** — its loss triggers *aneurysm*.
- **Collagen & SMC turn over constantly** — so the wall can *adapt*, or *run away*.

Teaching-grade values for a large elastic artery, in the range used by Humphrey, Cyron, Latorre and co-workers (order-of-magnitude, not a fit to one animal).

An Idealized Blood Vessel

A vessel is a **pressurised, perfused tube**. Two loads act on the wall: **blood pressure** P pushes it outward, and **blood flow** Q drags its inner surface. Cells sense both.



P lumen pressure $\rightarrow$ **circumferential stress** σ_θ Q volumetric flow $\rightarrow$ **wall shear stress** τ_w a inner radius, h wall thickness

Two Loads, Two Formulae: Laplace and Poiseuille

Balancing lumen pressure against wall tension (thin-walled tube) gives the **circumferential (hoop) stress**; steady laminar (Poiseuille) flow gives the **wall shear stress**:

Pressure load

$$\sigma_{\theta} = P \frac{a}{h} \quad (1)$$

Thicker wall $\Rightarrow$ lower stress; larger radius or pressure $\Rightarrow$ higher stress. The load the intramural cells (SMC, fibroblasts) feel.

Flow load

$$\tau_w = Q \frac{4\mu}{\pi a^3} \quad (2)$$

The endothelium senses τ_w ; note the steep a^{-3} dependence — a small radius change moves shear stress a lot.

Two Homeostatic Set-Points, Two Adaptations

Each stimulus has a preferred value the cells defend — the *set-points* $\sigma_{\theta,h}$ and $\tau_{w,h}$. Rearranging the two laws shows *what geometry restores each set-point*:

Pressure $\rightarrow$ thickness

Hold $\sigma_{\theta} = \sigma_{\theta,h}$:

$$h = \frac{P a}{\sigma_{\theta,h}} \propto P.$$

Higher pressure $\Rightarrow$ **thicker wall**.

Flow $\rightarrow$ radius

Hold $\tau_w = \tau_{w,h}$:

$$a = \left(\frac{4\mu Q}{\pi \tau_{w,h}} \right)^{1/3} \propto Q^{1/3}.$$

Higher flow $\Rightarrow$ **larger lumen**.

These two rules are the mechanobiology in miniature — and we can explore them analytically before writing any code.

Your turn: Pressure adaptation

Question. A patient becomes hypertensive: pressure rises by **50%** to $P' = 1.5 P$, but the radius a is held fixed. By how much must the wall thicken to bring the circumferential stress back to its set-point $\sigma_{\theta,h}$?

Resources. The Laplace law (1); homeostatic stress $\sigma_{\theta,h}$.

Hints. Write σ_{θ} before and after; set the after-value equal to $\sigma_{\theta,h}$ and solve for the new thickness h' .

✓ Answer: Pressure adaptation

Initially homeostatic: $\sigma_{\theta,h} = Pa/h$. After the step, demanding $\sigma_{\theta} = P'a/h' = \sigma_{\theta,h}$ with $P' = 1.5 P$ gives

$$h' = 1.5 h \quad (\text{a 50\% rise}). \quad (3)$$

Reading. The wall thickens *in proportion* to the pressure rise — exactly Grossman's observation, and the outcome a healthy hypertensive adaptation should reproduce. Radius unchanged, stress restored.

Your turn: Flow adaptation

Question. Chronically increased flow rises by **30%** to $Q' = 1.3Q$. By how much must the lumen radius change to restore the wall-shear-stress set-point $\tau_{w,h}$?

Resources. The wall-shear-stress law (2); set-point $\tau_{w,h}$.

Hints. Only a adapts. Set $\tau_w(Q', a') = \tau_{w,h}$ and use $\tau_{w,h} = 4\mu Q/(\pi a^3)$ to eliminate constants.

✓ Answer: Flow adaptation

From $\tau_{w,h} = 4\mu Q' / (\pi a'^3) = 4\mu Q / (\pi a^3)$, the radius ratio is $(a'/a)^3 = Q'/Q = 1.3$, so

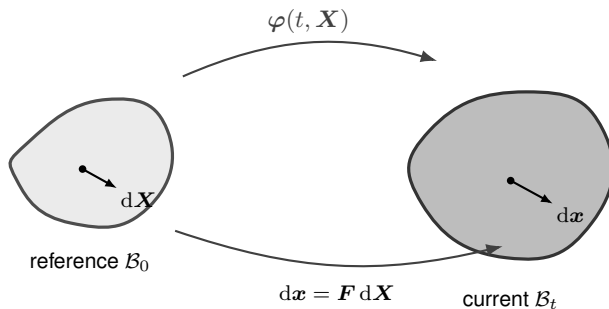
$$a' = 1.3^{1/3} a \approx 1.09 a \quad (\text{a 9\% rise}). \quad (4)$$

Reading. The lumen enlarges as the *cube root* of the flow increase — the classic flow-induced outward remodeling. A 30% flow rise widens the vessel by only ~9%.

Finite-strain fundamentals

Reference and Current Configuration

A body is described in two configurations: the **reference** $\mathcal{B}_0$ (material points $\mathbf{X}$) and the **current** $\mathcal{B}_t$ (spatial points $\mathbf{x}$). The **motion** $\mathbf{x} = \varphi(t, \mathbf{X})$ carries one to the other.



Our 0D Model vs. the 3D Geometric Model

Everything in this seminar has a **full 3D** form and a **reduced 0D** form. We use the *same notation* for both; the badges tell you which:

3D 3D geometric model

- full deformation field $\boldsymbol{x} = \boldsymbol{\varphi}(t, \boldsymbol{X})$
- tensor $\boldsymbol{F} = \partial\boldsymbol{\varphi}/\partial\boldsymbol{X}$; stresses $\boldsymbol{\sigma}, \boldsymbol{S}$
- solved on a meshed artery (finite elements)

0D 0D reduced model

- one dominant direction $\Rightarrow$ scalar stretch λ
- tensor $\boldsymbol{F} \rightarrow \lambda$; stress $\boldsymbol{\sigma} \rightarrow \sigma$
- a handful of ODEs / algebraic equations — runs in milliseconds

The 0D model is the thin-walled artery ($\sigma_\theta = Pa/h$) reduced to its one load-bearing direction — ideal for exploring the theories analytically.

The Deformation Gradient

How a line element transforms is the key kinematic quantity — the **deformation gradient** maps reference line elements to current ones.

3D general form — a tensor

$$\mathrm{d}\mathbf{x} = \mathbf{F} \mathrm{d}\mathbf{X}, \quad \mathbf{F} = \frac{\partial \boldsymbol{\varphi}}{\partial \mathbf{X}}, \quad J = \det \mathbf{F} \quad (5)$$

J is the volume ratio; **elastic incompressibility** means $J_e = 1$ (only growth changes volume).

0D reduction used here — one dominant direction

$$\mathbf{F} \longrightarrow \lambda = \frac{\text{current length}}{\text{reference length}}, \quad \lambda = 1 \text{ undeformed.}$$

Strain — A Rotation-Free Measure of Stretching

F still contains rigid rotation. Removing it gives the **right Cauchy–Green** tensor and the **Green–Lagrange strain**.

3D general form — tensors

$$\boxed{C = F^T F, \quad E = \frac{1}{2}(C - I)} \quad (6)$$

0D reduction used here — one dominant direction

$$C = \lambda^2, \quad E = \frac{1}{2}(\lambda^2 - 1).$$

Along the dominant loading direction the whole strain state collapses to a single stretch λ — all we need for the thin-walled vessel.

Stress — Force per Current Area

We report the **Cauchy (true) stress**: force per unit *current* area — the natural measure for a growing body, since the load-bearing area itself changes as the tissue remodels.

3D general form — a tensor (from the 2nd Piola–Kirchhoff S)

$$\boldsymbol{\sigma} = \frac{1}{J} \mathbf{F} \mathbf{S} \mathbf{F}^\top, \quad \mathbf{S} = 2 \frac{\partial W}{\partial \mathbf{C}}.$$

0D reduction used here — one scalar

$$\boldsymbol{\sigma} \longrightarrow \sigma \text{ (the value in Laplace, (1)).}$$

Balance of linear momentum in the current configuration reduces, for our quasi-static thin-walled vessel, to exactly that Laplace relation.

Hyperelasticity: Stress from a Stored-Energy Function

A **hyperelastic** material stores energy W ; its stress is a derivative of W — no dissipation, path-independent. Each constituent gets its own W : **elastin** is an isotropic *neo-Hookean* solid, **collagen** a *Fung-type* fiber.

3D general form — energy $W(\mathbf{C})$, stress $\mathbf{S} = 2 \partial W / \partial \mathbf{C}$

$$W_{\text{elastin}} = \frac{c}{2} (I_1 - 3), \quad W_{\text{collagen}} = \frac{c_1}{4c_2} \left[e^{c_2(I_4 - 1)^2} - 1 \right]$$

$I_1 = \text{tr } \mathbf{C}$ (isotropic elastin); I_4 reads $\mathbf{C}$ along the collagen fiber — the one place anisotropy enters.

0D reduction used here — one stretch λ_e , scalar $\sigma = \lambda_e dW/d\lambda_e$

$$\sigma_{\text{elastin}} = c (\lambda_e^2 - \lambda_e^{-1}), \quad \sigma_{\text{collagen}} = c_1 \lambda_e^2 (\lambda_e^2 - 1) e^{c_2(\lambda_e^2 - 1)^2} \quad (7)$$

The Deposition Stretch G^k

Cells deposit new fibers **already under tension** at a fixed deposition stretch $G^k > 1$. At homeostasis the elastic stretch equals G^k , which *defines* the homeostatic stress:

$$\sigma_h^k := \sigma^k(G^k) \quad (8)$$

No deposition pre-stress

New material enters slack ($\sigma_{\text{pre}} = 0$); as old fibers are removed, the replacement carries no load — the homeostatic state **cannot be maintained**.

With deposition pre-stress

New material enters at the homeostatic prestress $\sigma_{\text{pre}} = \sigma_h$ (stretch G^k); each replacement immediately bears its share — the state is **maintained through turnover**.

This one convention pins every set-point — and lets a continuously-renewed tissue hold a steady mechanical state.

Kinematic growth

Multiplicative Decomposition — Growth vs. Elasticity

Kinematic growth splits the deformation into a stress-free **growth** part F_g (how the tissue adds mass) and an **elastic** part F_e — the only part that carries stress.

3D general form — a product of tensors

$$\mathbf{F} = \mathbf{F}_e \mathbf{F}_g, \quad \text{only } \mathbf{F}_e \text{ stores energy.}$$

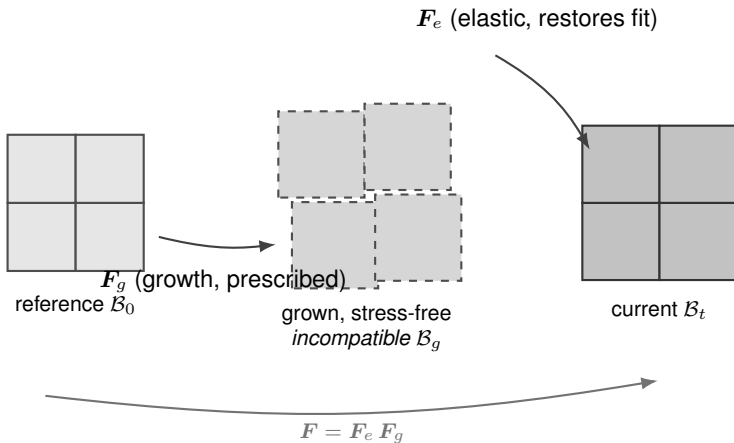
0D reduction used here — scalar stretches

$$\boxed{\lambda = \lambda_e \lambda_g} \tag{9}$$

Growth is *one* inelastic stretch λ_g ; the elastic stretch $\lambda_e = \lambda/\lambda_g$ is what the constituent laws see. (Constrained mixtures will later give each cohort its own natural configuration.)

The Idea, and the Kinematics

Postulate a stress-free **growth** deformation F_g ; only the elastic part F_e carries load. Grown pieces need not fit together — F_g leaves an *incompatible* stress-free state; F_e forces them back into one body (residual stress).



The Growth Tensor Must Be Prescribed

F_g is **chosen by the modeler**, not derived — it encodes *how* the tissue is assumed to add mass. A common choice grows the tissue along a single direction e_f (a wall-thickening or *fiber* direction):

3D prescribed growth tensor — this is an *input*

$$\boxed{\mathbf{F}_g = \mathbf{I} + (\theta - 1) \mathbf{e}_f \otimes \mathbf{e}_f = \text{diag}(1, 1, \theta)} \quad (10)$$

0D the one scalar that actually evolves

$$\theta = \frac{M}{M_0} \quad (\text{mass / volume ratio}), \quad \det \mathbf{F}_g = \theta.$$

Here e_f is the growth direction — it could be the radial (wall-thickening) direction, or a fiber direction. Everything about the growth *pattern* is fixed a priori; only the scalar **growth stretch** θ evolves in time.

The Growth Law

The growth *pattern* $\mathbf{F}_g$ is fixed; the growth law evolves its single scalar θ to null the stress deviation from the set-point (first-order, production $\propto$ current mass).

3D general form — rate of the growth tensor via its scalar

$$\dot{\mathbf{F}}_g = \dot{\theta} \mathbf{e}_f \otimes \mathbf{e}_f, \quad \dot{\theta} = k_g \theta \left(\frac{\bar{\sigma}}{\bar{\sigma}_h} - 1 \right).$$

0D scalar growth law — the only unknown in $\mathbf{F}_g$

$$\boxed{\frac{d\theta}{dt} = k_g \theta \left(\frac{\bar{\sigma}}{\bar{\sigma}_h} - 1 \right)} \quad (11)$$

The set-point $\bar{\sigma}_h$ is **prescribed**: wherever growth stops ($\dot{\theta} = 0$), stress is homeostatic *by construction*. At each instant λ follows from equilibrium $\bar{\sigma}(\lambda) = \sigma_{\text{req}}(\lambda, \theta)$.

Stability Is Imposed, Not Predicted

Is the homeostatic state θ^* stable? Perturb it, $\theta = \theta^* + \delta$, and linearise the growth law (11) about θ^* (where $\bar{\sigma} = \bar{\sigma}_h$):

$$\dot{\delta} = \underbrace{k_g \theta^* \frac{1}{\bar{\sigma}_h} \frac{d\bar{\sigma}}{d\theta} \bigg|_{\theta^*}}_{=: \alpha} \delta.$$

The perturbation decays ($\delta \rightarrow 0$, **stable**) iff $\alpha < 0$, i.e.

$$\left. \frac{d\bar{\sigma}}{d\theta} \right|_{\theta^*} < 0 \iff \text{adding mass } \textit{lowers} \text{ the stress it must carry.} \quad (12)$$

For the intact artery this holds, but a strong enough insult flips the sign (Laplace $\bar{\sigma} \propto \lambda^2$) — then $\alpha > 0$ and growth runs away. The set-point is *prescribed*, so the model can only *impose* stability, never *predict* it.

The `gr` Toolbox — Shared Setup for Every Coding Exercise

Every coding exercise reuses *one* problem definition, so the four theories are compared on identical numbers. A `Model` bundles the constituents with the homeostatic pressure $P_h = 13.3$ kPa (~ 100 mmHg) and reference mid-wall radius $R = 1$ mm; the reference thickness follows from the Laplace balance $H = P_h R / \bar{\sigma}_h$ (1).

Default arterial constituents (mass fraction ϕ_0 , deposition stretch G , law, turnover rate k_d , gain K_σ):

	ϕ_0	G	law, k_d [1/d], K_σ
elastin	0.30	1.40	neo-Hooke $c=90$; $k_d=0$
collagen	0.35	1.08	Fung $c_1=250$, $c_2=8$; $\frac{1}{20}$; 1
smc	0.35	1.20	Fung $c_1=120$, $c_2=3$; $\frac{1}{20}$; 1

Set-point per constituent $\sigma_h^k = \sigma^k(G^k)$; mixture $\bar{\sigma}_h = \sum_k \phi_0^k \sigma_h^k$.
Elastin never turns over ($k_d^e = 0$) and is the only *degradable* constituent (aneurysm).

Running an exercise. Each exercise is a small YAML *input file*; one runner drives them all — no Python to edit: `uv run python run.py configs/exNN_*.yaml`

The file picks the theory, geometry, insult and parameters; the run prints $\bar{\sigma}/\bar{\sigma}_h$, mass and diverged and plots the history. For a **parametric study**, list values in the file's `sweep`: block or on the command line: ... `-sweep`

`insult.pressure_factor=1.2,1.5,2.0` # one curve per value

... `-set model.constituents.collagen.gain=3.0` # change any one value

Geometry & equilibrium. At every step the supplied mixture stress must equal the Laplace required stress

$$\bar{\sigma}(\lambda) = \bar{\sigma}_h \gamma \frac{\lambda^2}{m},$$

solved for λ by a bracketed root find on $[0.2, 20]$; *no root* $\Rightarrow$ runaway. Here $m = M/M_0$ is the mass ratio, γ the load factor, and the exponent 2 is the Laplace term $\bar{\sigma} = Pa/h$.

Insult. Insult ramps a step from `t_on` over ramp days: `pressure_factor` γ (hypertension) or `elastin_surviving` s (aneurysm).

⚙️ Setup: Example — hypertensive artery (ex02)

Example. A pressurised artery is made *hypertensive* by a sustained +50% pressure step. Does the wall thicken back to its *prescribed* homeostatic stress (11), or run away once the pressure step is large enough?

Run it. `uv run python run.py configs/ex02_kinematic_growth.yaml`

The file sets theory: `kinematic_growth`, the insult and `k_g`. Watch $\bar{\sigma}/\bar{\sigma}_h$ return to 1 while θ climbs to the pressure factor.

Parametric study. Overlay several growth rates (the file's sweep: block), or push the pressure until it diverges: ... `-sweep simulate.k_g=0.02,0.05,0.2` # vary the gain

... `-set insult.pressure_factor=3.0` # harder insult

Your turn: Kinematic growth

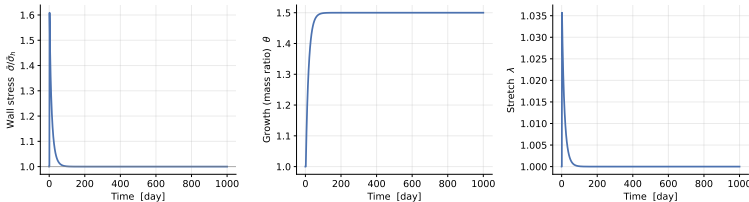
Question. Does the artery reach the *prescribed* homeostatic set-point — and how large a pressure step does it take to tip it into runaway instead?

Hints. Watch $\bar{\sigma}/\bar{\sigma}_h$ settle at 1 while θ climbs to the pressure factor. The Laplace λ^2 term is what lets the artery tip over once $d\bar{\sigma}/d\theta$ flips sign.

✓ Answer: Kinematic growth

The stress plateau *a/ways* lands on the prescribed set-point; the artery (Laplace λ^2) tips into runaway once PRESSURE_FACTOR is large enough — exactly when $d\bar{\sigma}/d\theta$ turns positive.

Kinematic growth of the artery under hypertension (P: 1.0 -> 1.5x)

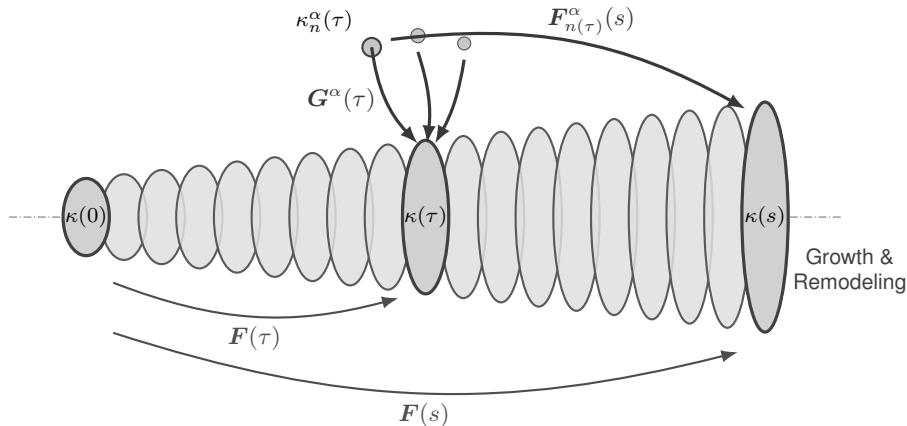


Learning: reaches the a-priori prescribed homeostasis, *or* it is unstable — nothing in between.

The constrained mixture model

The Idea: Track Every Cohort

Constituents are **constrained to deform together**, but each carries its *own* natural configuration and is continuously **produced and removed**. The tissue therefore carries a whole *history* of cohorts — faithful, and expensive.



Mass Balance — The Heredity Integral

The mass of a turnover constituent k now is the surviving remnant of the initial material plus everything produced since, each aged by a survival function:

$$M^k(t) = M^k(0) q^k(t, 0) + \int_0^t m^k(\tau) q^k(t, \tau) d\tau \quad (13)$$

Survival (Poisson removal at rate k_d^k):

$$q^k(t, \tau) = e^{-k_d^k(t-\tau)}.$$

The index d denotes *degradation*: k_d^k is the mass removal rate of constituent k , so $1/k_d^k$ is its mean lifetime; elastin does not turn over ($k_d^e = 0$).

Production $\propto$ current mass, driven by the tissue-stress stimulus that vanishes at homeostasis:

$$m^k = k_d^k M^k \Upsilon, \quad \Upsilon = 1 + K_\sigma^k \left(\frac{\bar{\sigma}}{\bar{\sigma}_h} - 1 \right).$$

Each Cohort Remembers Its Birth Configuration

A cohort of k deposited at time τ is laid down at deposition stretch G^k and thereafter stretches *with* the mixture.

3D general form — constituent-specific deformation (Humphrey Fig. 1)

$$\mathbf{F}_{n(\tau)}^k(t) = \mathbf{F}(t) \mathbf{F}^{-1}(\tau) \mathbf{G}^k.$$

0D reduction used here — scalar elastic stretch

$$\boxed{\lambda_e^k(t, \tau) = G^k \frac{\lambda(t)}{\lambda(\tau)}} \quad (14)$$

At birth ($t = \tau$) it sits exactly at G^k (hence σ_h^k). This is the **constrained-mixture constraint**: every constituent shares the current deformation, but each cohort measures it from a different starting point.

Mixture Stress = History Integral

The mixture Cauchy stress is the mass-weighted sum over *all* surviving cohorts (rule of mixtures):

$$\bar{\sigma}(t) = \frac{1}{M_{\text{tot}}} \sum_k \left[M^k(0) q^k(t, 0) \sigma^k(G^k \lambda(t)) + \int_0^t m^k(\tau) q^k(t, \tau) \sigma^k\left(G^k \frac{\lambda(t)}{\lambda(\tau)}\right) d\tau \right] \quad (15)$$

Each time step discretises the mass balance and this sum over stored cohorts, then solves $\bar{\sigma}(\lambda) = \sigma_{\text{req}}(\lambda)$ for the current stretch. Elastin contributes a single, non-renewing cohort (optionally degraded by an insult).

What It Buys You: Stability Is *Predicted*

Nothing prescribes the outcome — adaptation or dilation **emerges** from the turnover dynamics and the insult. Under hypertension the model grows collagen and muscle while elastin is diluted, and tissue stress returns to homeostatic on its own.

The price: the history integrals make the model $O(N^2)$ in time steps — the motivation for the homogenized model.

⚙️ Setup: Example — turnover under hypertension (ex03)

Example. The same artery, now resolving constituents that are continuously *produced and removed* (13) (elastin is not). Which constituent takes up the load, and does turnover restore the homeostatic stress? The same tissue may *adapt* to hypertension yet *run away* on an aneurysm insult.

Run it. `uv run python run.py configs/ex03_constrained_mixture.yaml`

The file sets each constituent's `turnover_days` ($1/k_d$) and `gain` (K_σ) and the insult. For the aneurysm, swap the insult to `elastin_surviving: 0.3`.

Parametric study. Sweep the pressure step (the file's `sweep: block`), or change a tissue property: ...

```
... -set model.constituents.collagen.gain=3.0
```

```
... -set model.constituents.collagen.turnover_days=70
```

Your turn: Constrained mixture

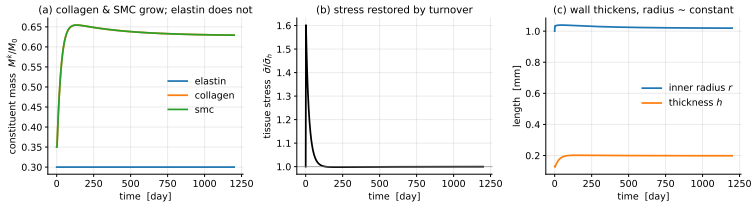
Question. Does the same tissue *adapt* to hypertension but *run away* on the aneurysm insult? Which constituent grows to carry the load elastin used to bear?

Hints. Try turnover 10 / 20 / 70 d and gain 0.3 / 1.0 / 3.0. Faster turnover and higher gain reach the new state sooner; watch which insult the tissue can absorb.

✓ Answer: Constrained mixture

Mild hypertension **adapts**: collagen and SMC grow, tissue stress returns to homeostatic. A severe elastin insult can **run away**. Faster turnover / higher gain reach the new state sooner.

Full constrained-mixture model: adaptation to hypertension (artery)



Learning: stability is *predicted*, and it **depends on the insult**.

Homogenized CMM

Throw Away the History, Keep Its Average

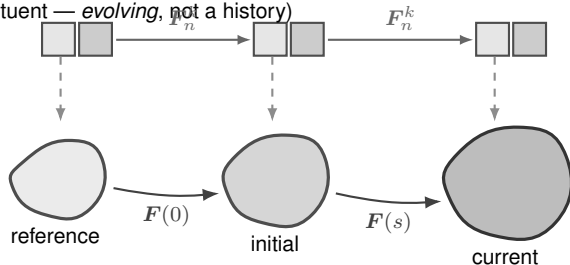
Temporal homogenization replaces the entire deposition history by a *single evolving mean natural configuration* per constituent — two ODEs per constituent, cost $O(N^2) \rightarrow O(N)$.

3D general form (Cyron 2016) — mean natural-config tensor

0D reduction — scalar mean stretch

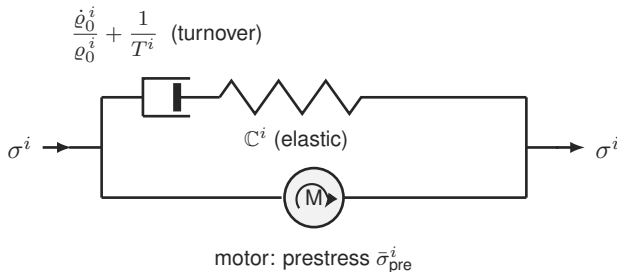
$$\mathbf{F}_e^k = \mathbf{F}(s) \mathbf{F}_n^k(s)^{-1} \quad \longrightarrow \quad \lambda_e^k = \frac{\lambda}{\lambda_n^k}.$$

stress-free *mean* natural configurations
(one per constituent — *evolving, not a history*)



The Mechanical Analog: A Maxwell Fluid with a Motor

Each constituent behaves like a **Maxwell element in parallel with a motor**: turnover relaxes stress toward a *prestress*, not zero.



σ^i	constituent stress (the load)
$\mathbb{C}^i$	elastic stiffness (spring)
$\dot{\varrho}_0^i / \varrho_0^i + 1/T^i$	turnover + degradation (dashpot)
$\bar{\sigma}_{\text{pre}}^i$	deposition prestress (motor)

Back to the full CMM (0D):

- spring $\leftrightarrow$ constituent law $\sigma^k(\lambda_e^k)$
- dashpot $\leftrightarrow$ mass turnover / removal k_d^k
- motor $\leftrightarrow$ deposition stretch G^k

Growth (Mass) and Remodeling (Natural Configuration)

Growth: mass production $\propto$ current mass, driven by stress deviation. **Remodeling:** the mean natural configuration evolves as new mass enters at λ/G^k and old mass leaves at the current value.

3D general form — mass & mean natural-config *tensor*

$$\dot{M}^k = M^k (K_\sigma^k k_d^k) \left(\frac{\bar{\sigma}}{\bar{\sigma}_h} - 1 \right), \quad \dot{\mathbf{F}}_n^k = f(\mathbf{F}, \mathbf{F}_n^k; m^k, M^k).$$

0D reduction — scalar mass and scalar natural stretch

$$\boxed{\frac{dM^k}{dt} = M^k K_\sigma^k k_d^k \left(\frac{\bar{\sigma}}{\bar{\sigma}_h} - 1 \right)} \quad (16)$$

$$\boxed{\frac{d\lambda_n^k}{dt} = \frac{m^k}{M^k} \left(\frac{\lambda}{G^k} - \lambda_n^k \right)} \quad (17)$$

Holding λ fixed, remodeling relaxes constituent stress exponentially to σ_h^k ($\sim 1/k_d$) — not to zero, but to a prestress.

It Tracks the Full Model Closely — Much Cheaper

Two ODEs per constituent replace the full deposition history. The homogenized trajectory sits almost on the full constrained mixture, while kinematic growth under-responds to elastin loss — it grows constituents in lockstep and cannot remodel their natural configurations individually.

The homogenized model follows the full model's *time trajectory* closely, at $O(N)$ instead of $O(N^2)$.

✂ Your turn: When do homogenized and full agree?

Question. The homogenized model keeps only the *mean* natural configuration per constituent. Under what assumptions should it match the full deposition-history model — and when would you expect them to diverge?

Resources. The two remodeling pictures: full heredity integral vs. the single mean stretch λ_n^k (17).

Hints. Think about the *spread* of natural configurations across cohorts. When is a single mean a good summary of the whole history?

✓ Answer: When do homogenized and full agree?

They agree when each constituent's cohorts share *nearly the same* natural configuration — i.e. slow, gradual loading where deposition stretch G^{tk} and turnover are roughly stationary, so the cohort spread stays narrow and the mean represents it well. **They diverge** under fast or strongly time-varying insults, when cohorts deposited at very different states coexist and a single mean stretch can no longer capture the distribution.

⚙️ Setup: Example — homogenized vs. full (ex04)

Example. Run both mixture theories on the *same* artery and insult. The homogenized model keeps one mean natural stretch per constituent (two ODEs, (16)–(17)) instead of the full cohort history. How closely does it track the full model, and how much faster is it?

Run it. `uv run python run.py configs/ex04_homogenized_vs_full.yaml`

Here study: compare runs theories: [constrained_mixture, homogenized_cmm] on one scenario and prints each runtime and the largest mass-ratio gap between them.

Parametric study. Stress the comparison with a harsher, longer insult: `... -set insult.elastin_surviving=0.3 -set insult.ramp=10`
`... -set simulate.t_end=4000 # does the agreement hold?`

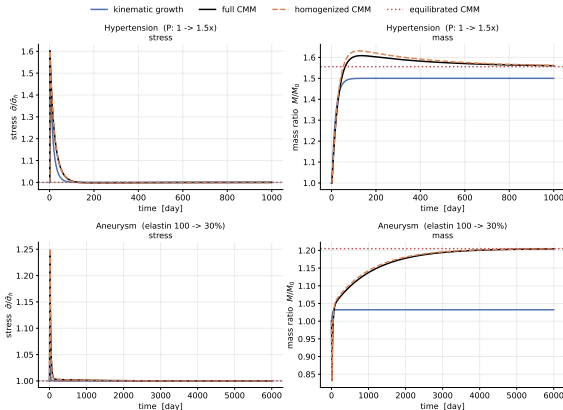
Your turn: Homogenized vs. full

Question. Is the difference between the two curves visible by eye? By what factor is the homogenized model faster — and does the agreement survive a harsher, longer insult?

Hints. Read the printed mass-ratio gap and the two runtimes. The homogenized model should track the full one closely wherever the loading is slow and gradual.

✓ Answer: Homogenized vs. full

The homogenized curve overlays the full CMM closely at a fraction of the runtime ($O(N)$ vs. $O(N^2)$). **Learning:** it reproduces a **similar time trajectory** to the full model — cheaply.



Equilibrated CMM

Skip the Transient — Solve for the End State

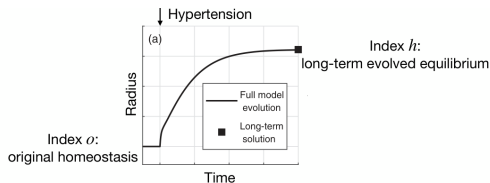
Set every rate to zero and solve the **algebraic** equilibrium — one root-find instead of thousands of time steps.

3D general form — the 3D equilibrium conditions

$$\underbrace{\Upsilon = 1 \Rightarrow \bar{\sigma}^* = \bar{\sigma}_h}_{\text{(A) growth balance}} \quad \underbrace{F_n^k = \text{const} \Rightarrow \sigma^k = \sigma_h^k}_{\text{(B) remodeling done}} \quad \underbrace{\nabla \cdot \sigma = 0}_{\text{(C) mechanical equilib.}}$$

0D reduction — scalar conditions at the evolved geometry

$$\bar{\sigma}^* = \bar{\sigma}_h, \quad \lambda_e^k = G^k, \quad \sigma_\theta = \frac{Pa}{h}.$$



Collapse to One Scalar Equation

Let the turnover constituents grow in fixed proportion (exact when they share gain and turnover). The 3D conditions (A)–(C) collapse to a *single* equation for the evolved stretch λ^* — consistent with the same λ , G^k , ϕ^k used throughout:

0D scalar equilibrium equation for λ^*

$$\phi^e s + s \frac{\bar{\sigma}_h - \sigma^e(G^e \lambda^*)}{\sigma_{\text{turn}} - \sigma_h^e} - \gamma (\lambda^*)^2 = 0 \quad (18)$$

with the Laplace exponent $n = 2$, γ the sustained load factor, s the surviving elastin fraction, ϕ^e the elastin mass fraction. Once λ^* is known, the evolved mass ratio is $M/M_0 = \gamma(\lambda^*)^2$.

This equation *is* the fixed point of the homogenized ODEs ((16)–(17) with $d/dt = 0$), so its root **coincides with the long-time limit of the transient theories** — to several digits.

The Punchline: Existence = Stability

Equation (18) may have **no physical root** — then no adapted state exists and the transient theories dilate without bound. The solver is therefore the cleanest **stability test**: root $\Rightarrow$ adapts, no root $\Rightarrow$ unstable. Near the boundary the transient slows critically, which is exactly why the instant algebraic test is so useful.

⚙️ Setup: Example — the equilibrated stability test (ex05)

Example. Skip the transient: solve the single algebraic equation (18) for the *final* adapted stretch λ^* . A physical root means the tissue adapts; *no root* means it dilates without bound. Scanning elastin loss then traces the stability boundary.

Run it. `uv run python run.py configs/ex05_equilibrated.yaml`

Here `study: equilibrium` solves once for the given insult, then the `scan: block` sweeps `elastin_surviving` and `report_critical: elastin` prints the critical surviving fraction s_{crit} .

Parametric study. Move the boundary by stiffening the wall, then re-read s_{crit} : ... `-set model.constituents.collagen.gain=3.0`
... `-set model.constituents.collagen.law.c1=600`

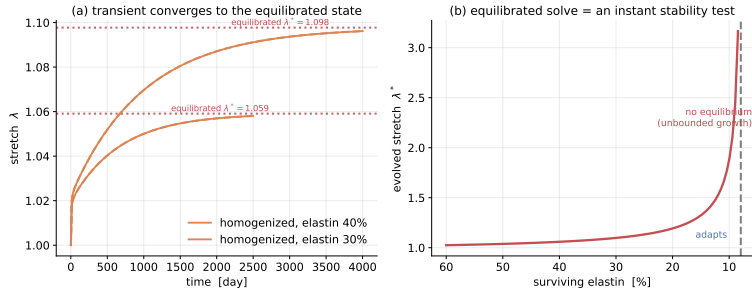
Your turn: Equilibrated

Question. Does the algebraic λ^* land on the transient plateau where a root exists — and what is the critical elastin loss beyond which *no* equilibrium exists at all?

Hints. The scan curve shoots up as $s \rightarrow s_{\text{crit}}$: the adapted stretch diverges just before the root vanishes. Below s_{crit} the transient models dilate without bound.

✓ Answer: Equilibrated

Where a root exists, the equilibrated solve lands exactly on the transient end-state. Below s_{crit} there is *no* root — the transient dilates without bound.



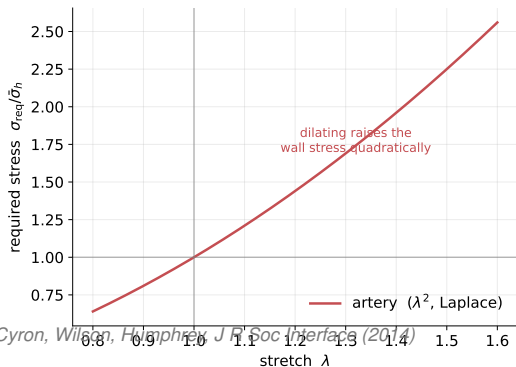
Learning: matches the transient theories *if* an equilibrium exists — and flags when it does not.

Stability

Where Instability Comes From: The Laplace Exponent

In a pressurised tube the wall stress is $\bar{\sigma} = \frac{Pa}{h}$ with radius $a \propto \lambda$ and thickness $h \propto M$, so the required (Laplace) stress is *quadratic* in the stretch, $\bar{\sigma}_{\text{req}} \propto \gamma \lambda^2 / M$. The homeostatic state is stable only if adding mass *lowers* the stress it must carry:

$$\left. \frac{d\bar{\sigma}}{d\theta} \right|_{\theta^*} < 0 \iff \text{tissue adapts} \quad (19)$$



Growth thickens the wall ($h \propto M$), which lowers stress; but dilation raises it as λ^2 (Laplace).

While enough **elastin** survives, its stiff recoil keeps $d\bar{\sigma}/d\theta < 0$ and the artery **adapts**. Lose enough elastin and the sign flips, $d\bar{\sigma}/d\theta > 0$ — **runaway**. It is the quadratic Laplace exponent that makes this possible.

Same Tissue, Two Insults

Read stability two ways: **existence** — the equilibrated equation (18) has a root iff the tissue can adapt (no time integration); or **transient** — integrate and watch it plateau or climb. The same tissue can do either: mild elastin loss adapts, severe loss dilates without bound.

⚙️ Setup: Example — the stable/unstable map (ex06)

Example. A tissue *adapts* to an insult iff a mechanobiological equilibrium exists (18). Sweeping that instant test over (surviving elastin) $\times$ (pressure) draws the stable/unstable map. Then change *one* tissue property and watch the unstable “aneurysm” region shrink.

Run it. `uv run python run.py configs/ex06_stability.yaml`

Here study: map builds the 2D grid over the two axes: via the adapts existence test and shades adapt (blue) vs. runaway (red); it also prints the fraction of the grid that adapts.

Parametric study. Change one property in the `model`: block (gain, collagen `c1`, or G) — e.g.

```
... -set model.constituents.collagen.gain=3.0
```

```
... -set model.constituents.collagen.G=1.15
```

— and compare how much the stable region grows.

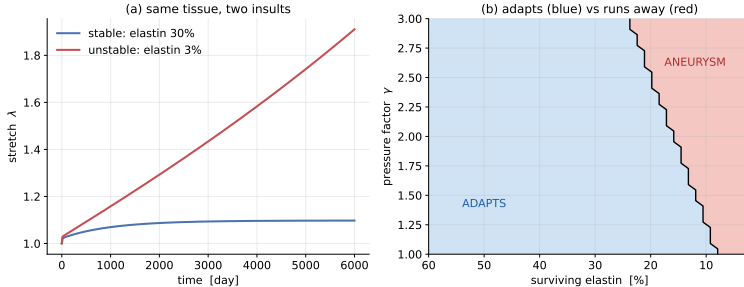
Your turn: Stability map

Question. Which single change — higher gain, stiffer collagen, or larger deposition stretch — enlarges the stable region the most? Elastin cannot be regrown in adults: what does that imply for aneurysm?

Hints. Compare the printed “fraction that adapts” across the three changes. Each shifts the boundary between adaptation and runaway.

✓ Answer: Stability map

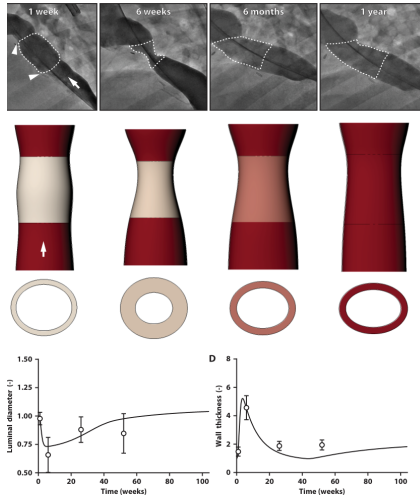
Higher gain, stiffer collagen, or larger deposition stretch all enlarge the *adapts* region — the tissue tolerates a stronger insult before running away.



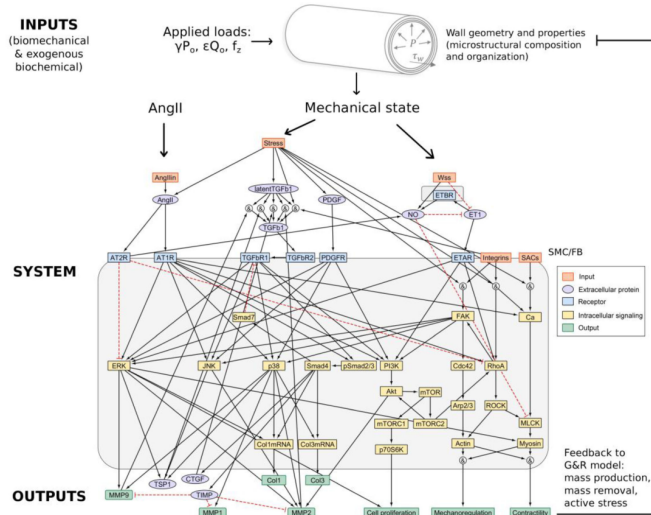
Learning: the existence test that defines the equilibrium boundary *is* the stability criterion.

Future perspectives

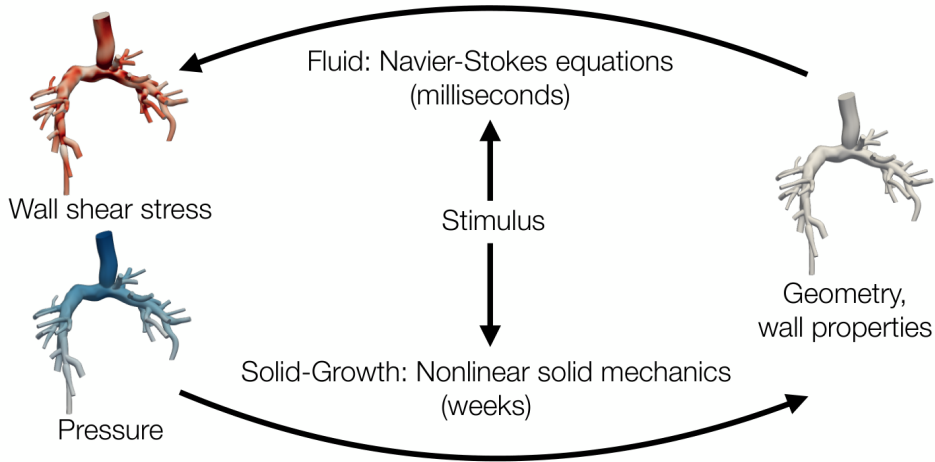
Clinical Application: Reversible Graft Stenosis



Systems-Biology Coupling: Signaling to Remodeling



Fluid–Solid–Growth: Closing the Hemodynamic Loop

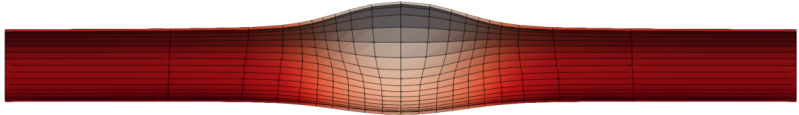


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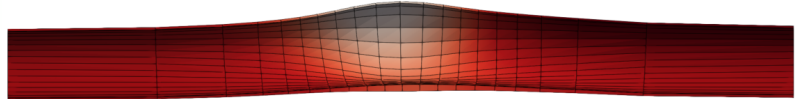
FSGe: Long-Term Evolved Aneurysmal Equilibrium

Solid with $\tau_w \sim 1/r^3$

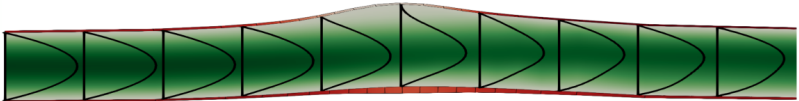
Wall shear
stress



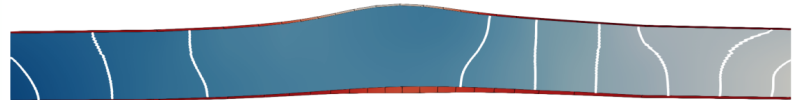
**Fluid-Solid-Growth
interaction**



Velocity



Pressure



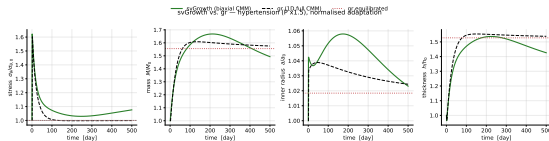
Open-source implementations

This Seminar's Code: A Reproducible Workflow

Everything you have seen is generated by the `gr` Python package in **one repository**:

- all four theories on *one* parameter set and *one* mechanical setting — apples-to-apples;
- deliberately reduced-order (1D / thin-membrane, single stress component) so it runs in seconds;
- docs/ lecture notes, configs/ YAML input files driven by one `run.py` (edit a file or pass `-sweep` — no Python to touch), `solutions/` that regenerate every figure.

make figures in `slides/` regenerates fig01–06 from the code, so the deck never drifts from the model.



Research-Grade Open-Source Solvers

Project	What it does	Stack / group
svGrowth	constrained-mixture G&R of vessels (biaxial thin-wall); the cross-validation reference used above	Python; Stanford CBCL
svFSGe	fast, strongly-coupled 3D fluid–solid–growth driver (the FSGe method)	Python + svFSI/svMultiPhysics; Stanford CBCL
FEMBE_Plugin	mechanobiologically equilibrated constrained-mixture material as a FEBio plugin	C++ / FEBio; Yale (Humphrey lab)
FEFSG_Plugin	fluid–solid–growth coupling as a FEBio plugin (thoracic-aorta hypertension & aneurysm cases)	C++ / FEBio; Yale (Humphrey lab)

github.com/StanfordCBCL/svGrowth github.com/StanfordCBCL/svFSGe

github.com/yale-humphrey-lab/FEMBE_Plugin github.com/yale-humphrey-lab/FEFSG_Plugin

How the Four Theories Answer — The Takeaways

Theory	Stability?	What it says
Kinematic growth	<i>Imposed</i>	Either reaches the <i>a-priori prescribed</i> homeostasis, or is unstable — nothing in between. Built in, not learned.
Constrained mixture (full)	<i>Predicted</i>	Adapts or dilates depending on the insult — the faithful reference.
Homogenized CMM	<i>Predicted</i>	A similar time trajectory to the full model, at a fraction of the cost.
Equilibrated CMM	<i>Instantly</i>	Matches the transient theories <i>if</i> an equilibrium exists; <i>no</i> root $\Rightarrow$ it does not match — unbounded growth.

Stability *depends on the insult*, and — with turnover-based theories — it can be *predicted*, not just assumed.

Summary — What to Take Away

The biology. Living tissue continuously **turns over**; cells deposit new material *pre-stretched* at G^k , chasing a homeostatic mechanical set-point σ_h .

The four theories, one set-point, one question — does it adapt or run away?

- **Kinematic growth** — prescribe F_g ; *imposes* the set-point. Outcome is binary: reach the prescribed homeostasis, or go unstable.
- **Constrained mixture** — track every cohort; stability is *predicted* and **depends on the insult** (mild $\rightarrow$ adapt, severe $\rightarrow$ runaway).
- **Homogenized** — one mean natural configuration per constituent; a **similar time trajectory** to the full model at $O(N)$ not $O(N^2)$.
- **Equilibrated** — solve for the end state directly; **matches** the transient theories *iff* an equilibrium exists, and flags instability by finding no root.

Where it goes. These tools now drive **clinical prediction** (TEVGs), **systems-biology coupling**, and **fluid–solid–growth**.

When does a loaded tissue adapt, and when does it run away? — now you can compute the answer.

Key references I

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Key references II

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Key references III

Code

- Seminar code (this deck's figures): `github.com/YaleCBL-Teaching/Growth-Remodeling`
- `github.com/StanfordCBCL/svGrowth`, `svFSGe`
- `github.com/yale-humphrey-lab/FEMBE_Plugin`, `FEFSG_Plugin`

Figure sources. Result figures reproduced with the `gr` package; literature figures cited per-slide.